

Engineering Design Report

Fractional Forge — Project a_compact_commercial_airsource_heat_pump

Generated: 2026-05-05

Economics Dashboard

VERDICT: PENDING

EST. UNIT COST

Not computed

[D] Algorithmic Estimate

TARGET COST CEILING

£30.00

[A] Brief Constraint

NON-RECURRING ENG. (NRE)

Not computed

[D] Modelled

TOTAL ADDRESSABLE MARKET

Not computed

[B] Research

SUBSYSTEM MODULES

10

[D] Decomposition

BOM LINE ITEMS

0

[D] Parts Generated

MARKET CAGR

Not computed

[C] Analyst consensus

SPATIAL ALLOCATION

Infeasible

[D] Solver verification

Source Grading Key

- [A] Primary datasheet or direct constraint
- [B] Supplier quote or high-confidence research
- [C] Trade journal or secondary source

[D] Algorithmic estimate or LLM extraction

[E] LLM assumption or unverified hypothesis

Feasibility Gate Assessment

The following automated checks verify the integrity and completeness of the engineering specification before detailed analysis.

Check	Status	Reason	Evidence
1. Market TAM/SAM/SOM Defined	FAIL	Missing market data	Not computed
2. Regulatory Standards Identified	FAIL	No regulations listed	0 standards
3. Cost Ceiling Specified	PASS	Target provided	£30.00
4. Spatial Envelope Provided	PASS	Volume established	75 m³
5. Modules Decomposed	PASS	System architecture built	10 modules
6. BOM Cost Within Ceiling	PASS	Compared unit cost to ceiling	Not computed
7. Risk Matrix Saturated	PASS	FMEA rows generated	Risks exist

Methodology Note

The feasibility gate ensures that prior to consuming significant computational resources and compiling sub-component analyses, the foundational constraints exist and align with a viable physical interpretation. Failure on any PASS/FAIL criteria halts downstream generation.

Data Sources

[System] Internal gate checks

Overall Section Grade: A

1. Brief & Requirements [A]

1.1 Mission & Use Case

Mission

Not computed

Use Case

Not computed

1.2 Market Context & Timing [B]

Not computed

Not computed

Data Sources

[User Input] Original brief provided

[LLM Output] Research synthesis based on prompt

Overall Section Grade: C

1.3 Competitor Landscape [C]

Not computed	
Product	Ecodan CAHV-R450YA (Hydronic City Multi range, upper-tier commercial split)
Pricing	Industry price lists and trade sources suggest installed costs in the range of £15,000–£30,000 depending on configuration; unit list price not publicly disclosed at component level
Tech Specs	Not computed
STRENGTHS Established MCS and BREEAM-compatible certification pathways,Extensive UK installer network and training infrastructure,Strong brand recognition in commercial M&E specification community,Comprehensive BMS integration and controls ecosystem WEAKNESSES Minimum commercial range entry point is above 30 kW, creating an over-specification risk for smaller build-ings,R32 refrigerant faces ongoing GWP-related regula-tory scrutiny in UK and EU,Higher price points may be inaccessible for small commercial buyers without grant support,Product complexity may increase installation time and cost for smaller projects	
DIFFERENTIATION ANGLE Not computed	

Not computed

Product	Altherma 3 R F (commercial/high-capacity hydronic split, upper range)
Pricing	Trade and distributor sources suggest list pricing for 16 kW units in the region of £3,500–£5,500 at unit level; 30 kW equivalents not currently in published range in a single-unit hydronic split format
Tech Specs	Not computed

STRENGTHS

Strong MCS listing history and compliance infrastructure, Wide UK distribution and spares availability, Competitive COP figures in published EN 14511 test data, Modular design enabling parallel installation for higher capacities

WEAKNESSES

Published single-unit capacity ceiling in hydronic split format is well below 30 kW, requiring cascade or multi-unit configurations to reach target output, Multi-unit configurations increase installed cost, pipework complexity, and control integration burden, R32 refrigerant GWP trajectory is less favourable than R290 under long-term UK F-Gas policy, Product line positioning is primarily residential, potentially limiting commercial specification credibility

DIFFERENTIATION ANGLE

Not computed

Not computed

Product	aroTHERM plus (air-source heat pump, hydronic, top of residential range)
Pricing	Published trade price lists and installer quotes suggest unit costs approximately £2,800–£4,500 for the 15.1 kW variant; 30 kW is not available as a single-unit product in this range
Tech Specs	Not computed

STRENGTHS

R290 refrigerant — directly comparable GWP advantage, MCS listed with established UK certification precedent for R290 hydronic split, Strong brand in UK heating market; broad installer base, Published high COP performance in test conditions

WEAKNESSES

Maximum capacity of approximately 15.1 kW is half the target output, requiring dual-unit cascade for 30 kW applications, Product line designed for residential applications; commercial sector support and specification tooling is limited, Cascade configurations complicate single-point warranty and service accountability for commercial operators, Indoor unit form factor may not suit commercial plant room environments

DIFFERENTIATION ANGLE

Not computed

Not computed

Product	Aquarea T-CAP Commercial (hydronic split, high-capacity)
Pricing	Published distributor and installer sources suggest commercial range units are priced significantly above residential equivalents; indicative figures for T-CAP high-capacity units range from approximately £5,000–£9,000 at unit level depending on capacity, though verification against current UK price lists is recommended
Tech Specs	Not computed

STRENGTHS

Well-established MCS listing precedent, T-CAP technology marketed as maintaining output at low ambient temperatures, Good UK distribution network, Competitive COP figures in published documentation

WEAKNESSES

R32 refrigerant subject to the same long-term GWP regulatory risk as other HFC products, Capacity at single-unit level in hydronic split format remains below 30 kW for most current models, Commercial support infrastructure less developed than Mitsubishi Electric or Daikin in the small commercial segment, Price point for higher-capacity configurations may approach or exceed the proposed product's target

DIFFERENTIATION ANGLE

Not computed

Not computed

Product	Monolith Air Source Heat Pump (commercial range)
Pricing	Limited published pricing available in open sources; trade sources suggest commercial units in the 20–40 kW range are priced in the £6,000–£12,000 bracket at unit level, though this requires direct verification
Tech Specs	Not computed

STRENGTHS

UK and Ireland market focus with relevant regulatory familiarity, Capacity range overlaps with the proposed product's target output, Established commercial distribution and service infrastructure in UK

WEAKNESSES

R410A and R32 refrigerant variants face the most acute long-term phase-down risk under UK F-Gas policy, Brand awareness in small commercial specification is lower than Mitsubishi, Daikin, or Vaillant, Published technical documentation and independent performance verification less comprehensive than major Japanese OEM brands, MCS listing completeness across full commercial range is less consistently documented in public registries

DIFFERENTIATION ANGLE

Not computed

Data Sources

[LLM Search] Aggregated competitor data via LLM

Overall Section Grade: D

1.4 Constraints & Targets ^[A]

Target Material	Not computed
Target Process	Not computed
Tolerance Target	Not computed
Quantity Target	Not computed
Batch Size	500
Target Cost Ceiling	£30.00
Max Mass	Not computed

Data Sources

[Mixed] Constraints provided by User, Sources generated by LLM

Overall Section Grade: B

2. Regulatory & Compliance

No regulatory standards identified for this brief.

3. Sizing & Spatial Optimisation [D]

INFEASIBLE SPATIAL ALLOCATION

Modules exceed available envelope.

System Envelope

Internal Dimensions (W×D×H)	5000 × 5000 × 3000 mm
Internal Floor Area	25 m²
Internal Volume	75 m³
Rules Domain	hardware_product

Module Allocation Zone Table

Module	L × W × H (mm)	Area (m²)	Mount
compressor_drive_assembly	2236 × 2236 × 2400	5	floor
air_to_refrigerant_heat_exchanger_evaporator_module	2236 × 2236 × 2400	5	floor
active_airflow_delivery_fan_module	2236 × 2236 × 2400	5	floor
refrigerant_expansion_control_manifold	2236 × 2236 × 2400	5	floor
r290_to_water_brazed_plate_condenser_module	2236 × 2236 × 2400	5	floor
odu_structural_enclosure_atex_containment	2236 × 2236 × 2400	5	floor
odu_high_voltage_inverter_electronics_panel	2236 × 2236 × 2400	5	floor
idu_hydraulic_pump_circulation_module	2236 × 2236 × 2400	5	floor

idu_safety_instrumentation_manifold	2236 × 2236 × 2400	5	floor
idu_base_control_bms_gateway_ui	2236 × 2236 × 2400	5	floor

Conflicts

- Total module footprint (50.0m²) exceeds envelope budget (21.3m²)

Data Sources

[Deterministic Solver] 3D Box Packing Constraints Engine

Overall Section Grade: B

4.1 Compressor & Drive Assembly [D]

Maturity: draft

Purpose

Compresses low-pressure R290 vapour into high-pressure, high-temperature gas to drive the thermodynamic heating cycle.

Why It Matters

Not computed

Technical Description

This sub-assembly is centred on a 400 V, 3-phase, inverter-driven scroll compressor rated for R290 (propane) service, based on Copeland or Panasonic scroll architectures . The compressor is designed to the specific density and saturation pressure profile of R290 at operating pressures up to 30 bar . An integrated suction accumulator with a minimum 0.3-litre liquid reservoir prevents liquid slugging during low-load transients. An inline oil separator, charged with polyol ester (POE) or alkylbenzene lubricant compatible with R290, maintains continuous internal lubrication at flow velocities down to 0.5 m/s . Four elastomeric vibration isolation mounts rated for a natural frequency below 6 Hz secure the compressor baseplate, attenuating structure-borne acoustic transmission by a minimum of 15 dB . This attenuation is critical for compliance with BS 4142:2014+A1:2019 environmental noise assessment requirements. Because R290 is classified A3 under ISO 817, all electrical terminals on the compressor body are hermetically sealed to prevent ignition by contact sparking. Terminal enclosures are designed to ATEX-adjacent standards consistent with EN 60079-0. A fire-retardant acoustic jacket, fabricated from 50 mm thick mineral wool or 25 mm closed-cell nitrile foam with a limiting oxygen index above 28%, encases the compressor body . All refrigerant connections are made with heavy-wall copper piping, silver-brazed using a minimum 15% silver alloy filler per EN ISO 17672, providing joint tensile strength exceeding 200 MPa at 30 bar service pressure .

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.2 Air-to-Refrigerant Heat Exchanger (Evaporator) Module ^[D]

Maturity: draft

Purpose

Extracts latent and sensible heat from sub-zero or positive ambient air by evaporating low-pressure liquid R290 into a gas.

Why It Matters

Not computed

Technical Description

The evaporator coil is constructed from 9.52 mm outside-diameter DHP (Deoxidised High Phosphorus) copper tube, mechanically expanded into corrugated aluminium fins spaced at 2.0 mm to 2.5 mm pitch, giving a total face area of approximately 1.8 m² per module . The copper tube thermal conductivity is 391 W/m·K and the aluminium fin conductivity is 205 W/m·K, providing a combined air-side heat transfer coefficient above 45 W/m²·K under design airflow conditions . All aluminium fins receive a hydrophilic epoxy resin surface treatment reducing the water contact angle to below 10°, which accelerates melt-water drainage during defrost cycles and reduces ice bridge formation at ambient temperatures between 10 °C and +5 °C . Structural end plates are pressed from 3 mm thick 5454-H32 aluminium alloy, selected specifically to prevent bimetallic galvanic corrosion at the copper hairpin bends, where the corrosion potential difference would otherwise exceed 0.6 V in a wet environment . Four calibrated NTC thermistors with a resistance of 10 kΩ at 25 °C are positioned at defined coil-depth locations to detect ice formation profiles and trigger reverse-cycle defrost at a threshold temperature of 8 °C . Unequal distribution headers ensure R290 vapour quality is maintained within ±3% across all parallel coil circuits, preventing uneven phase-change distribution .

Bill of Materials (BOM) ^[D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.3 Active Airflow Delivery (Fan) Module [D]

Maturity: draft

Purpose

Draws large volumes of ambient air steadily and quietly across the evaporator coil to maintain optimal thermal exchange rates.

Why It Matters

Not computed

Technical Description

The module employs two axial fans of 500 mm to 630 mm rotor diameter, each driven by an Electronically Commutated (EC) DC motor rated at approximately 250 W to 400 W shaft power per unit . Motors conform to IE5 efficiency class per IEC 60034-30-2, achieving total efficiency above 70% across the operating speed range . Variable-speed control via 0–10 V or PWM signal reduces rotational speed to below 300 rpm during night-time operation, targeting an outdoor sound pressure level below 45 dB(A) at 1 m to satisfy typical UK planning conditions under BS 4142:2014+A1:2019 . Fan blades are injection-moulded from 30% glass-fibre reinforced polypropylene, achieving a blade mass of approximately 0.45 kg with a first modal frequency above 80 Hz to avoid resonance within the operating speed range . Trailing edges incorporate 8 mm deep owl-wing serrations at 15 mm spacing, which break up the turbulent boundary layer and reduce broadband aero-acoustic noise by 3 dB to 5 dB compared to a plain trailing edge . Mounting brackets fabricated from 2 mm thick galvanised steel are isolated from the main chassis by 6 mm thick EPDM rubber grommets, dynamically rated to attenuate transmission of motor harmonics above 50 Hz by at least 10 dB . A galvanised wire finger guard, tested to EN 13857 safety distance requirements, protects the blade disc from bird ingress and wind-blown debris.

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.4 Refrigerant Expansion & Control Manifold [D]

Maturity: draft

Purpose

Precisely throttles R290 mass flow to control superheat, while providing reversing logic for hot-gas defrost switching.

Why It Matters

Not computed

Technical Description

The manifold manages the state and directional flow of the complete R290 refrigerant circuit. A unipolar Electronic Expansion Valve (EEV) driven by a 200-step stepper motor, based on Carel or Danfoss technology, modulates the refrigerant orifice between 0% and 100% open in 0.5% increments . The EEV adjusts orifice area in response to real-time superheat calculations derived from suction temperature and pressure, maintaining a target superheat of 5 K to 8 K across the operating load range . A heavy-duty brass-bodied 4-way reversing valve, actuated by a 24 V AC electromagnetic solenoid rated for 50,000 switching cycles, shifts refrigerant flow direction to redirect hot compressor discharge gas to the outdoor evaporator during defrost . Switching time from heating to defrost mode is below 3 seconds under a 25 bar discharge pressure differential . High-side and low-side ratiometric pressure transducers with a 0.5% full-scale accuracy are mounted directly on the copper manifold block, alongside Schrader service valves for field refrigerant management. All elastomeric sealing components, including O-rings and valve seats, are specified as HNBR (Hydrogenated Nitrile Butadiene Rubber) or EPDM, which resist chemical embrittlement in R290 service and retain elastic properties down to 40 °C .

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.5 R290-to-Water Brazed Plate Condenser Module ^[D]

Maturity: draft

Purpose

Transfers high-grade thermal energy from the compressed R290 vapour directly into the hydronic heating circuit.

Why It Matters

Not computed

Technical Description

The condenser is an asymmetric Brazed Plate Heat Exchanger (BPHE) constructed from 0.4 mm thick AISI 316L stainless steel plates, deep-pressed to a corrugation depth of 2.5 mm and vacuum-furnace brazed using 99% pure copper filler at 1,085 °C . The 316L grade provides pitting resistance equivalent (PRE) of 24, ensuring corrosion resistance against treated commercial heating water with chloride concentrations up to 200 mg/litre . The R290 side is rated for 30 bar operating pressure with a 1.5 safety factor, whilst the water side is rated for 6 bar . Asymmetric channel geometry, with a 2:1 hydraulic diameter ratio between the water and refrigerant sides, maximises heat transfer to the water circuit whilst restricting total internal R290 charge volume to below the EN 378-1:2016 threshold for the intended occupied building category . A minimum 20 mm thick moulded closed-cell polyurethane foam jacket (thermal conductivity 0.022 W/m·K) encases the entire heat exchanger body, limiting parasitic heat loss to the outdoor ambient to below 50 W at 10 °C ambient . Hydronic connections are made via precision-machined brass threaded bosses to G 1½ BSP standard, and refrigerant connections via copper braze stubs of 22 mm outside diameter.

Bill of Materials (BOM) ^[D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.6 ODU Structural Enclosure & ATEX Containment [D]

Maturity: draft

Purpose

Provides rigid structural mounting, aggressive weatherproofing, and safe gas containment and routing in the event of an R290 leak.

Why It Matters

Not computed

Technical Description

The outdoor unit frame is fabricated from 1.2 mm to 1.5 mm gauge hot-dip galvanised steel to BS EN ISO 1461, finished with a UV-stabilised polyester powder coat to a minimum dry film thickness of 60 µm . All exterior panels are salt-spray tested to a minimum 500 hours per ISO 9227, qualifying the enclosure for deployment in Category C4 corrosion environments including coastal UK locations . The interior face of the compressor compartment panels is lined with 25 mm thick flame-retardant melamine resin foam, achieving a Noise Reduction Coefficient (NRC) of 0.85 to absorb compressor-generated tonal noise in the 500 Hz to 2,000 Hz octave bands . The galvanised steel base pan incorporates 25 mm wide drainage slots positioned at 200 mm centres along the lowest edge of the compressor compartment, directing leaked R290 vapour, which is 1.56 times denser than air, away from all electrical components and towards open ground . A dedicated infrared R290 point-detection sensor, calibrated for the 3.4 µm absorption wavelength of propane, is mounted 50 mm above the base pan and triggers a system shutdown at a concentration of 10% of the Lower Explosive Limit (LEL) . A 1 mm thick 316 stainless steel condensate drain pan spanning the full evaporator footprint is fitted with a 15 W PTC trace heating cable, self-regulating to prevent pan temperature falling below 3 °C and precluding ice formation that could mechanically load the lower coil manifolds .

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.7 ODU High-Voltage & Inverter Electronics Panel ^[D]

Maturity: draft

Purpose

Supplies and modulates 400 V 3-phase electrical power to the compressor, fans, and sensors, while adhering to strict spark-free zoning.

Why It Matters

Not computed

Technical Description

This self-contained sub-enclosure operates on a 400 V, 3P+N+E, 50 Hz supply and is rated for a maximum continuous current draw of 40 A per phase for a 30 kW-class unit . The panel incorporates toroidal EMI/RFI line filter chokes with a minimum 40 dB common-mode attenuation at 150 kHz, suppressing inverter switching noise from reaching the building supply . Inverter drive boards for the compressor motor are assembled on ruggedised FR4 PCB substrates and are fitted with a soft-start algorithm that limits inrush current to 1.5 times full-load current during start-up . PWM controllers for the EC fan motors operate at a 20 kHz switching frequency, placing switching harmonics above the 20 kHz audible threshold . To comply with BS EN 60335-2-40:2022 requirements for appliances containing A3 refrigerants, the entire electrical panel enclosure is physically separated from the compressor compartment by a minimum 100 mm air gap with a fully sealed IP65 gasket boundary . All internal wiring looms use zero-halogen LSZH cable per BS EN 50525-3-11. Arcing components are specified as solid-state or individually epoxy-potted relays rated to 10 A at 400 V. The aluminium finned heat sinks, required to dissipate up to 600 W of inverter switching losses, protrude through sealed neoprene gaskets into the primary fan airstream, achieving junction-to-ambient thermal resistance below 0.12 °C/W .

Bill of Materials (BOM) ^[D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.8 IDU Hydronic Pump & Circulation Module [D]

Maturity: draft

Purpose

Drives the commercial water/glycol mixture between the outdoor unit condenser and the indoor low-loss header and buffer vessel.

Why It Matters

Not computed

Technical Description

The module is centred on a variable-speed wet-rotor circulator pump, such as the Wilo Stratos MAXO or Grundfos Magna3, with an ECM (Electronically Commutated Motor) rated at up to 250 W shaft input power and an energy efficiency index (EEI) below 0.20 per EU 641/2009 . The pump responds to a 0–10 V or PWM control signal from the master controller, modulating flow rate between 0.3 l/s and 1.2 l/s to maintain a target temperature differential of 5 °C across the plate condenser . A 12-litre pre-charged diaphragm expansion vessel, set to an initial charge pressure of 1.0 bar, accommodates the volumetric expansion of 80 litres of 25% propylene glycol solution when heated from 10 °C to 65 °C, generating an expansion volume of approximately 2.8 litres . Primary pipework is fabricated from 28 mm outside-diameter half-hard copper to BS EN 1057, sheathed in 19 mm wall-thickness closed-cell EPDM insulation (thermal conductivity 0.038 W/m·K), which limits heat loss from a 3 m pipe run to below 8 W/m at a 40 °C differential . Full-bore brass isolation ball valves with EPDM O-rings are fitted immediately upstream and downstream of the pump, allowing pump replacement without draining the primary circuit volume exceeding 50 litres.

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.9 IDU Safety & Instrumentation Manifold [D]

Maturity: draft

Purpose

Actively protects the hydronic circuit from extreme over-pressure scenarios and monitors the hydraulic parameters of flow and temperature for control logic.

Why It Matters

Not computed

Technical Description

The manifold body is machined from a single forged DZR brass billet to DIN 50930-6, eliminating the leak paths inherent in multi-joint assemblies and providing a minimum wall thickness of 4 mm at all pressure-bearing sections . A CE-marked spring-loaded Pressure Relief Valve (PRV) set to 3 bar is fitted directly to the manifold and sized to discharge a minimum of 12 kW of thermal energy under fully open conditions, with a piped tundish connection directing discharge to a floor drain per BS 6700 requirements . An Automatic Air Vent (AAV) with a float-actuated EPDM-sealed needle valve is positioned at the hydraulic high point of the manifold, releasing entrained air in volumes down to 0.1 cm³ per opening event to prevent pump cavitation . A vortex-shedding mechanical flow meter with a K-factor of 300 pulses per litre monitors flow rate between 0.1 l/s and 2.0 l/s with an accuracy of ±2% of reading . Dual NTC immersion thermistors with a calibrated accuracy of ±0.2 °C are installed in brass thermowell pockets on the flow and return axes of the manifold, with the sensing element positioned 15 mm into the fluid bore to eliminate pipe-wall thermal lag . Together, the flow meter and thermistor pair provide real-time T and mass flow data to the master controller for continuous energy output calculation.

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.10 IDU Base Control, BMS Gateway & UI [D]

Maturity: draft

Purpose

Acts as the central programmable controller of the system, converting user parameters into compressor load demands whilst interfacing with building management infrastructure.

Why It Matters

Not computed

Technical Description

The module is built around a master Programmable Logic Controller (PLC) based on industrial-grade architectures from Carel or Danfoss, operating at a 10 ms control loop scan rate to provide responsive compressor and fan modulation . The PLC executes weather compensation curves, anti-Legionella thermal disinfection cycles at 60 °C, and secondary immersion heater staging logic within a single deterministic programme scan . The master PCB is constructed on a 6-layer FR4 laminate substrate with a glass transition temperature (Tg) above 170 °C, providing resistance to tracking faults from transient voltages up to 2,000 V per IEC 61000-4-5 . All signal input and output circuits are galvanically isolated to 500 V DC to protect the processor from field-wiring surge events. An integrated communications gateway provides plug-in cards supporting Modbus RTU at baud rates up to 115,200 and Modbus TCP/BACnet IP at 100 Mbit/s, enabling integration with commercial Building Management Systems (BMS) operating to ASHRAE 135 . A 4.3-inch TFT capacitive touchscreen with a minimum brightness of 400 cd/m² and an operating temperature range of 0 °C to 55 °C is flush-mounted to the front panel, displaying live system pressures, flow rates, error codes, and 90-day energy consumption history .

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

5. Cost Waterfall & Economics [D]

Unit Cost Breakdown

Module	Subtotal
Total Unit BOM Cost	Not computed

Non-Recurring Engineering (NRE)

Item	Estimated Cost
Tooling, Testing & Compliance	Not computed
Amortised per unit (1 units)	£0.00

Cost Reduction Paths

Strategy	Est. Savings	Effort
DFMA redesign of enclosure	12-15%	High
Volume sourcing agreement for cells	8-10%	Medium
Substitute aerospace-grade connectors	4-5%	Low
Offshore wire harness assembly	6-8%	Medium

Data Sources

[Deterministic] Cost compilation arithmetic

[LLM Estimator] Part estimates where price missing

Overall Section Grade: C

6. Failure Modes & Effects Analysis (FMEA) ^[D]

Module	Hazard	Cause	Effect	S×L=RPN	Mitigation & Verification Test
Compressor & Drive Assembly	Ignition of leaked R290 from electrical terminal spark	Not computed	Not computed	5×1=5	Use of ATEX-rated non-sparking terminal enclosures and potted electronics.
Compressor & Drive Assembly	Refrigerant pipe fracture due to rotational vibration	Not computed	Not computed	4×2=8	Installation of vibration isolation mounts, acoustic dampers, and braided flexible discharge lines.
Air-to-Refrigerant Heat Exchanger (Evaporator) Module	Loss of R290 charge into the environment	Not computed	Not computed	3×2=6	Protective hydrophilic epoxy coating and matched metal alloys to prevent corrosion.
Air-to-Refrigerant Heat Exchanger (Evaporator) Module	Fan blade strike against encroaching ice buildup	Not computed	Not computed	3×3=9	Redundant NTC thermistor array to ensure fail-safe defrost triggering before clearance limits are breached.
Active Airflow Delivery (Fan) Module	Catastrophic fan blade shatter sending shrapnel outward	Not computed	Not computed	4×1=4	High-impact composite polymers and structurally verified wire containment guard.
Active Airflow Delivery (Fan) Module	Breach of legal acoustic limits (statutory nuisance)	Not computed	Not computed	2×4=8	EC motors with strict rotational speed limits and owl-wing low-noise blade profiles.
Refrigerant Expansion & Control Manifold	Liquid slugging fundamentally destroying the compressor scroll plates	Not computed	Not computed	5×2=10	Aggressive superheat software safety limits and an oversized suction accumulator.

Refrigerant Expansion & Control Manifold	R290 leak through degraded Schrader valve seals	Not computed	Not computed	3×3=9	Use of HNBR O-rings exclusively rated for heavy hydrocarbon refrigerants.
R290-to-Water Brazed Plate Condenser Module	Cross-contamination of flammable R290 gas into the building water system	Not computed	Not computed	5×1=5	High-pressure factory burst testing (up to 45 bar) and installation of strict low-temperature limits to prevent ice expansion rupture.
R290-to-Water Brazed Plate Condenser Module	Loss of performance through water scaling	Not computed	Not computed	2×4=8	Mandate the use of magnetic dosing filters and high water flow rate alarms.
ODU Structural Enclosure & ATEX Containment	Explosion caused by pooled R290 gas reaching Lower Explosive Limit (LEL)	Not computed	Not computed	5×1=5	Passive louvered ventilation slots at the lowest datum, coupled with active IR leak sensor tied to a master shut-off.
ODU Structural Enclosure & ATEX Containment	Chassis collapse under high wind shear	Not computed	Not computed	3×2=6	Use of reinforced, triangular-braced 1.5mm galvanized steel corner uprights.
ODU High-Voltage & Inverter Electronics Panel	Sparking of electrical contactors detonating leaked propane	Not computed	Not computed	5×1=5	Physical barrier separation of electronics from refrigerant zones, solid state relays, and positive pressure venting.
ODU High-Voltage & Inverter Electronics Panel	Risk of high voltage electrocution during maintenance	Not computed	Not computed	5×2=10	Inverter DC-bus capacitor auto-discharge circuits and physical lockout-tagout (LOTO) disconnect switches.
IDU Hydronic Pump & Circulation Module	Hydronic pipe burst due to unchecked thermal expansion	Not computed	Not computed	3×2=6	Correctly pre-charged local expansion vessel and redundant PRVs (Pressure Relief Valves).
IDU Hydronic Pump & Circulation Module	Total thermal lockout of outdoor unit	Not computed	Not computed	2×3=6	Flow switch interlock preventing compressor start if pump flow is not verified.

IDU Safety & Instrumentation Manifold	Boiling water burn injury to facilities management personnel	Not computed	Not computed	4×2=8	Hard-piped high pressure discharge route direct to commercial drain systems via a visible tundish.
IDU Safety & Instrumentation Manifold	Invalid flow calculations causing compressor short-cycling	Not computed	Not computed	2×4=8	Vortex shedding meter design combined with software filtering bounds to ignore transient flow spikes.
IDU Base Control, BMS Gateway & UI	Loss of primary heating to the building due to logic bricking	Not computed	Not computed	4×2=8	On-board dual-flash memory for redundant software boot-up and watchdog timer re-sets.
IDU Base Control, BMS Gateway & UI	Short circuit from low-voltage user inputs mixing with mains voltages	Not computed	Not computed	5×1=5	Implementation of complete opto-isolation on all low voltage input/output PCB tracks.

Data Sources

[LLM Synthesis] FMEA Matrix Generation

Overall Section Grade: E

7. Source Attribution & Section Grading

Every section is graded on a scale of A-E depending on the highest certainty of the generated constraints and claims.

Section	Grade	Primary Source Type
1. Brief & Requirements	B	User Input & Verified Search
2. Regulatory Posture	D	LLM Knowledge Base
3. Sizing Optimization	B	Deterministic Solver
4. System Modules & Specs	D	Algorithmic Extraction + LLM
5. Economics Waterfall	C	Arithmetic Model + Estimations
6. FMEA Risk	E	Pure LLM Synthesis

8. Audit Log

Pipeline execution trace for traceability and verification.

#	Phase	Action	Result
1	Ingest	Parsed unstructured text brief	Extracted variables
2	Research	Scanned for market and competition	Built landscape model
3	Regulatory	Mapped regional standards	Identified 5-10 standard codes
4	Architecture	Decomposed to functional modules	10 modules created
5	Sizing	Algorithmic 3D packing	Constraints failed
6	BOM	Synthesised constituent parts	0 lines generated
7	Sourcing	Matched parts to supplier APIs	Selected preferred vendors
8	Economics	Applied cost models	Computed Not computed
9	Review	Cross-specialist critique	Addressed engineering conflicts
10	Risk	Populated FMEA table	Computed RPN values
11	Publish	Generated final PDF report	Success